hypethesis

iSCSI

What is it?
iSCSI (Internet SCSI) adapts the widely used SCSI standard for connecting hard drives to computers over a network connection such as a LAN or even the Internet. By carrying SCSI commands over IP networks, it is used to facilitate data transfers over intranets and to manage storage over long distances.

How does it work?
When a user sends a request, the operating system generates the appropriate SCSI commands and data request. IP packets are transmitted over an Ethernet connection. When a packet is received, it is separated into the SCSI commands and the request. The commands are sent to the SCSI controller and then to the SCSI storage device. Because iSCSI is bi-directional, it can also be used to return data in response to the original request.

What is its implication?
iSCSI has just been formally adopted and standardised. It builds over existing infrastructure, using SCSI hard disks, Ethernet networks and the Internet. Because of the ubiquity of IP networks, iSCSI can be used to transmit data over local area networks, wide area networks, or the Internet and can enable location-independent data storage and retrieval.

First JPEG, now MP3

The Fraunhofer Institute, developer of the MP3 codec, has announced that it will start charging licensing fees from users of the MP3 decoding algorithm (previously it charged only for the encoders). Red Hat has already removed all software packages from the next Linux release that would fall under the licensing scheme proposed by the Institute.

The fees of $0.75 (approx Rs 37) per decoder would add up to $16.5 million (approx Rs 82.5 crore) for downloaders of Winamp from Download.com alone. Fraunhofer also allows a one-time fee of $60,000 (approx Rs 30 lakh) for unlimited use of the decoder. Encoders and users of the mp3PRO software are taxed more heavily, with a levy of $2.50-$5 (approx Rs 125-250) per encoder. Use of the mp3PRO decoder software plus patent rights requires $1.25 (approx Rs 62) per unit or $90,000 (approx Rs 40.5 lakh) up front.

ILLUSTRATIONS: Mahesh Benkar

A consequence of these fees has been a resurgence in interest in competing free audio codecs. The Ogg Vorbis codec is a patent-free, professional audio codec that has been gaining steam recently. In an open letter to Thomson Multimedia, creators of the MP3 licensing scheme, Vorbis.com has stated its unabated glee at their actions, thanking them for the free publicity and noting that traffic to its site has increased substantially since the revocation of the free-licensing status of the Fraunhofer codec.

Power-up Mac

Apple revealed souped-up Power Macs in the first major upgrade to the professional system in about a year and a half. The new Power Macs sport faster memory and speedier system architecture. Apple also reintroduced dual-processor systems. Processor speeds range from 867 MHz to 1.25 GHz.

The new systems come with a 167 MHz system bus, but are still well behind a 533 MHz bus for the fastest Pentium 4 systems. Greg Joswiak, Apple's senior director, defends, "Our architecture is tough to compare to a PC bus, because we don't run everything off the same bus the way a PC does."

The changes also include a move to 266 MHz DDR SRAM and Mac OS X 10.2 preinstalled. The biggest incentive for purchasing a Power Mac would be the dual-processor architecture—Mac processors max out at 1.25 GHz, compared to more than double that for PCs. Support for a second optical drive and up to four hard drives complete the circle.

The 1 GHz system comes with an 80 GB hard drive, DVD recording drive and 64 MB ATi Radeon 9000 Pro.

Linux wins

In New York's Hudson Valley region, IBM's fabrication plant will be the only one that, when fully deployed, will produce chips using all three of the industry's bleeding edge technologies: low-k dielectrics, copper interconnect and silicon-on-insulator-based transistors. IBM will build chips on 12-inch wafers that pass from station to station in wafer pods on centrally controlled, automated, elevated rails. To run this, Linux was evaluated against a Windows-based system and performed flawlessly for three months, whereas the Windows-based system failed after six or seven days.

■ WorldCom's former CFO Scott Sullivan denies fraud, bailed at $500,000 ■ Napster faces liquidation ■ Microsoft, with others, campaigns against open source